

CoRoT-2b

R-0255

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_230626 · created 2026-07-17T07:33:07+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460121.9004 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.173 +/- 0.061
Transit depth (Rp/Rs)^2	3.0 +/- 2.11 [%]
Orbital Inclination (inc)	89.37 +/- 2.02
Airmass coefficient 1 (a1)	3.98 +/- 0.33
Airmass coefficient 2 (a2)	-1.11 +/- 0.11
Scatter in the residuals of the lightcurve fit is	8.02 %
Best Comparison Star	#2 - [275, 206]
Optimal Aperture	12.99
Optimal Annulus	32.47
Transit Duration (day)	0.0953 +/- 0.006

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.173 $\pm$ 0.061 vs 0.1667 (deviation 0.1 σ)
Duration fit vs published	0.0953 d vs 0.095 d (deviation 0.05 σ)
Mid-time O-C	8.7 min
Depth SNR	2.8
Residual scatter	8.02 %

Light curve

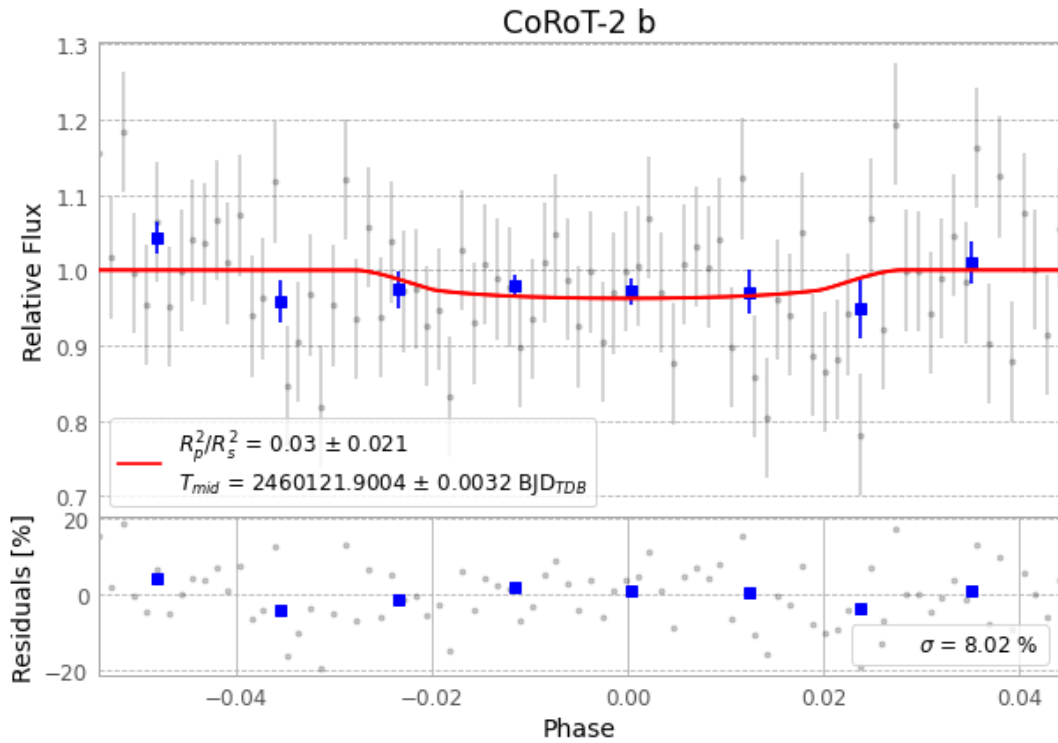


Figure 1 — FinalLightCurve_CoRoT-2 b_2023-06-26.png (SHA-256 ebd4225d7ee41b3b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [82, 346], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 517908d7ca9682a8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

83 FITS frames (55 MB), CoRoT-2230626071216.FITS → CoRoT-2230626111815.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-230626.log (6f379a7bb6c9...), FinalLightCurve_CoRoT-2 b_2023-06-26.png (ebd4225d7ee4...), FinalLightCurve_CoRoT-2 b_2023-06-26.csv (9b7e71c9e1aa...)

Compute resources

Wall time 140.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 07:33Z.